

Can Blood Vessel Networks in Ultrafast Breast MRI Predict Breast Cancer Treatment Response?

Predicting contrast changes improved the GNN, but it may also
be learning differences in scan timing

Spencer Venancio

with Sarit Bose and Aakrithi Ram · Advisor: Anna Woodard, PhD

Neoadjuvant therapy is given before surgery, and can shrink a tumor.

Treatment: Neoadjuvant therapy is given before surgery to shrink the tumor.

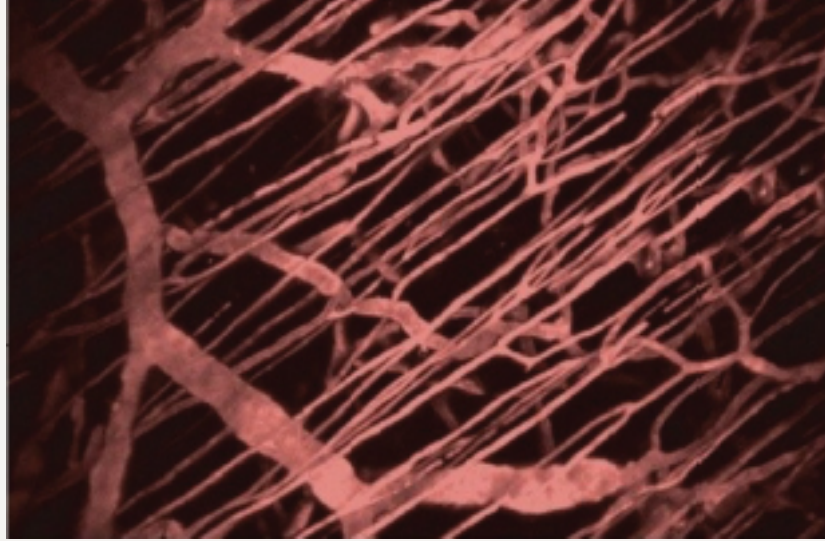
Outcome: About 1 in 3 exams came from patients who achieved a pathologic complete response (pCR): no invasive cancer remained in the breast or nearby lymph nodes at surgery. Achieving pCR is associated with better long-term outcomes.

Question: Can a GNN use connections within breast vessel networks to improve treatment-response prediction?

A graph neural network, or GNN, learns from both measurements at vessel locations and the connections between them.

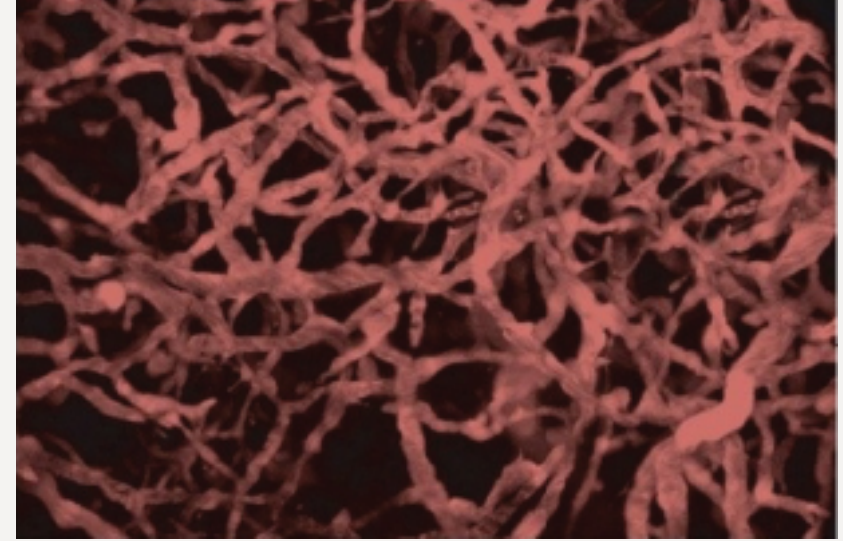
Earlier prediction could guide less intensive therapy for patients likely to achieve pCR and alternative therapy for those unlikely to do so.

Tumors build a messier blood supply



Normal vessels: organized, efficient.

Blood flows through a predictable, well-structured network.

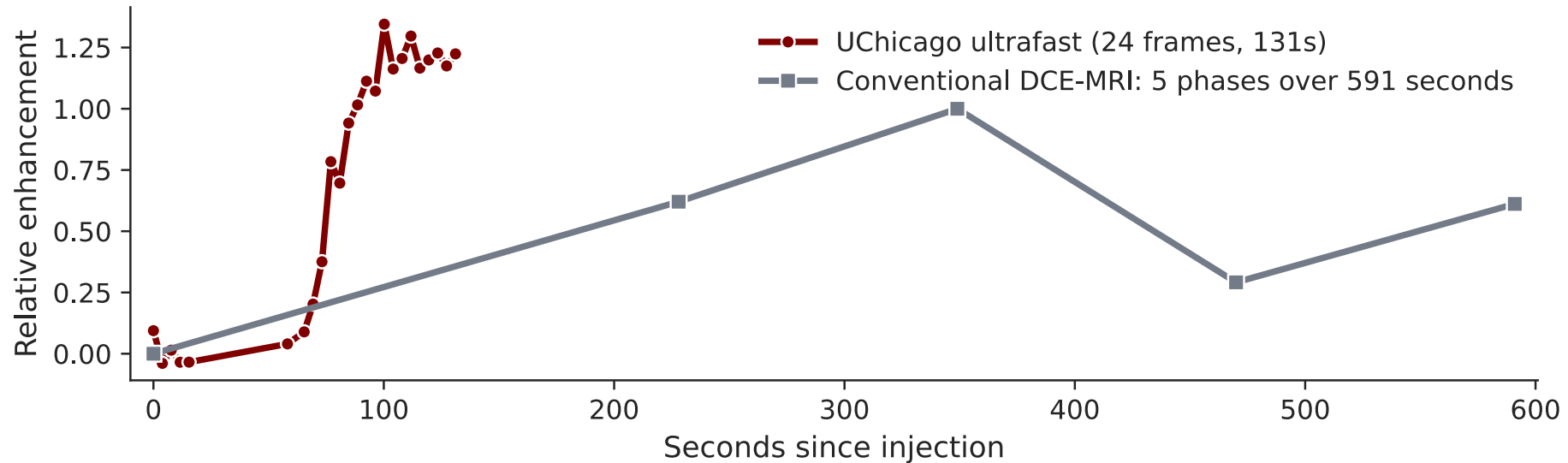


Tumor vessels: dense, tortuous, leaky.

Tumors stimulate abnormal vessel growth as they grow. Structure may reflect how aggressive a tumor is.

Because tumor vessels differ in both structure and blood flow, those patterns may help predict treatment response.

Ultrafast MRI movie shows more than anatomy



Curves are from different patients and are shown only to compare temporal sampling.

179

exams with known
outcomes

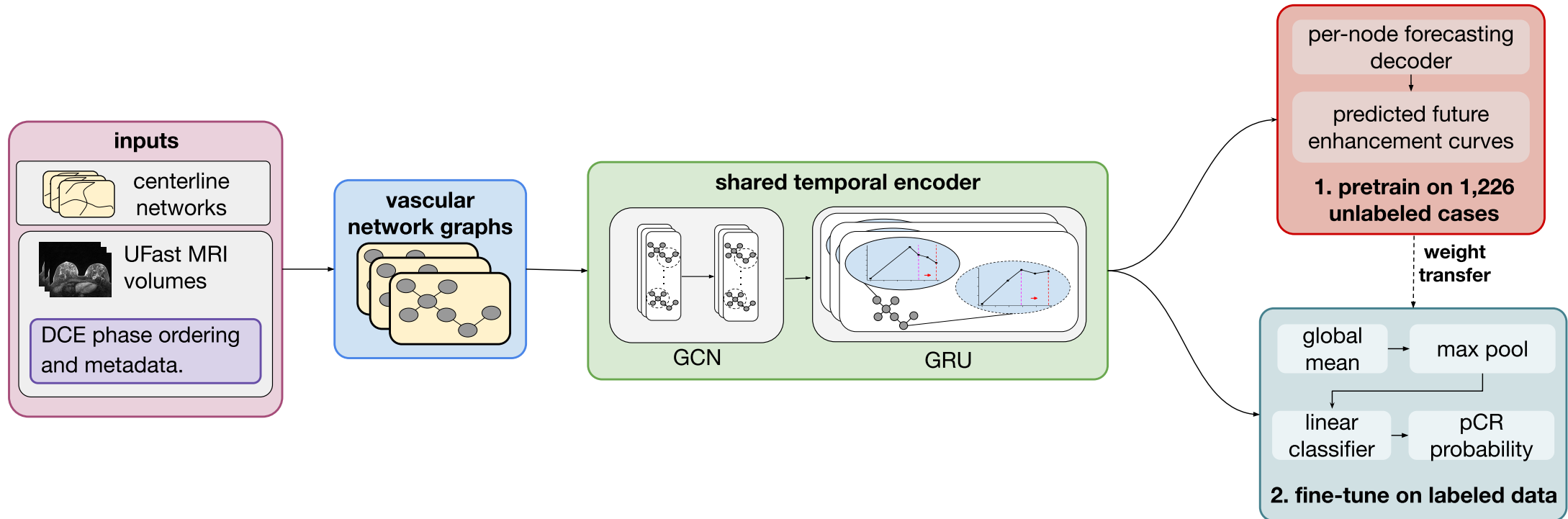
1,226

additional exams without outcome labels, used to teach
the model how contrast changes over time

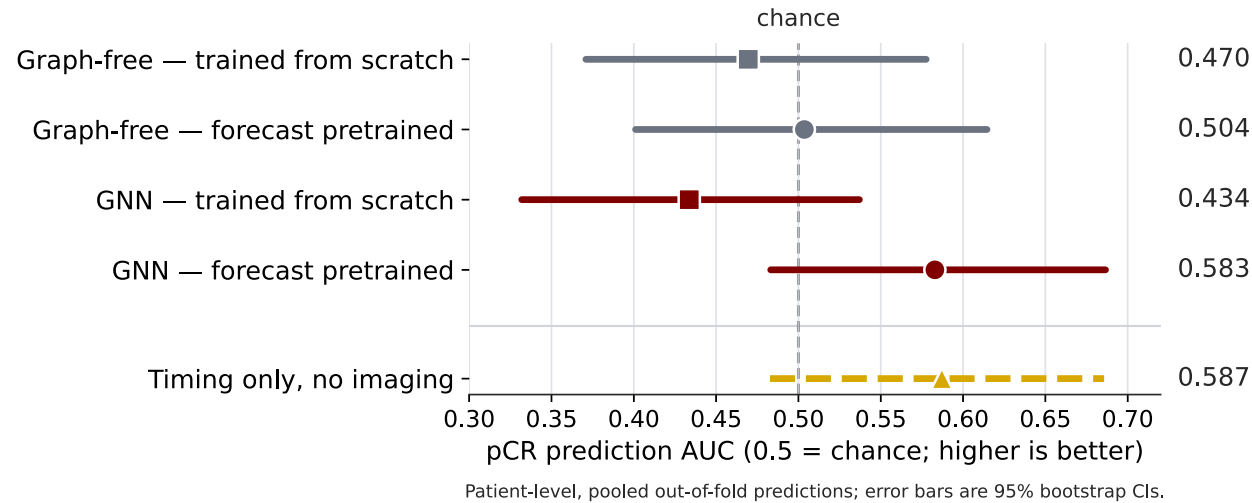
13–24

frames per ultrafast movie

Does the graph structure help? How about pretraining?



Learning to predict later MRI frames improved the GNN



Pretraining helped.

Learning to predict later MRI frames improved the GNN by **0.149 AUC** (95% CI, 0.046–0.255). After pretraining, the GNN also outperformed the matched graph-free model by **0.079** (95% CI, 0.010–0.155).

But scan timing was associated with pCR too.

A model using only scan-timing information, no imaging, reached **AUC 0.587** (95% CI, 0.483–0.686), and its predictions correlated with the GNN's at **$r = 0.50$** . We don't yet know why; next we need to determine what drives that association and whether it influences the GNN's predictions.

What should you remember?

1

Learning to predict later MRI frames improved the **GNN**, which then outperformed the matched graph-free model.

2

Enhancement measurements without vessel connections **remained near chance.**

3

How the scans were timed also predicted pCR, so we need to learn whether the model is detecting **tumor biology or scan differences.**

Learning to predict contrast changes improved the GNN, but some of its signal may come from how the scans were collected.

Thank you

Questions welcome.

Code

github.com/dsi-clinic/vanguard

Contact

svenancio@wisc.edu

Thanks to Zhen Ren, Milica Medved, and Greg Karczmar for feedback and MRI-physics expertise; Fred Howard for the development cohort; and Bella Summe, Julia Luo, José Cardona Arias, and Rebecca Wu for earlier vessel-pipeline development.